KP16 RBF Networks

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.15

Slides: read and download

Learning objectives

- Write the form of an RBF network and explain local against global response.
- Solve XOR with two centres and relate the line in the feature space to the closed curve in the original space.
- Explain how centres are chosen, the dimension price of locality and the contrast between the MLP and the RBF.

KP16 RBF Networks

One-sentence summary: Replace the interpolation function by a function of the distance to a centre so that every node responds only to data near itself; locality makes it learn fast and also forces it to fill the space with many centres, hungry for dimensions.

1. Gaussian nodes

Plain explanation: The products of KP15 respond over the whole space. Can the new coordinates be nodes that respond only to data nearby?

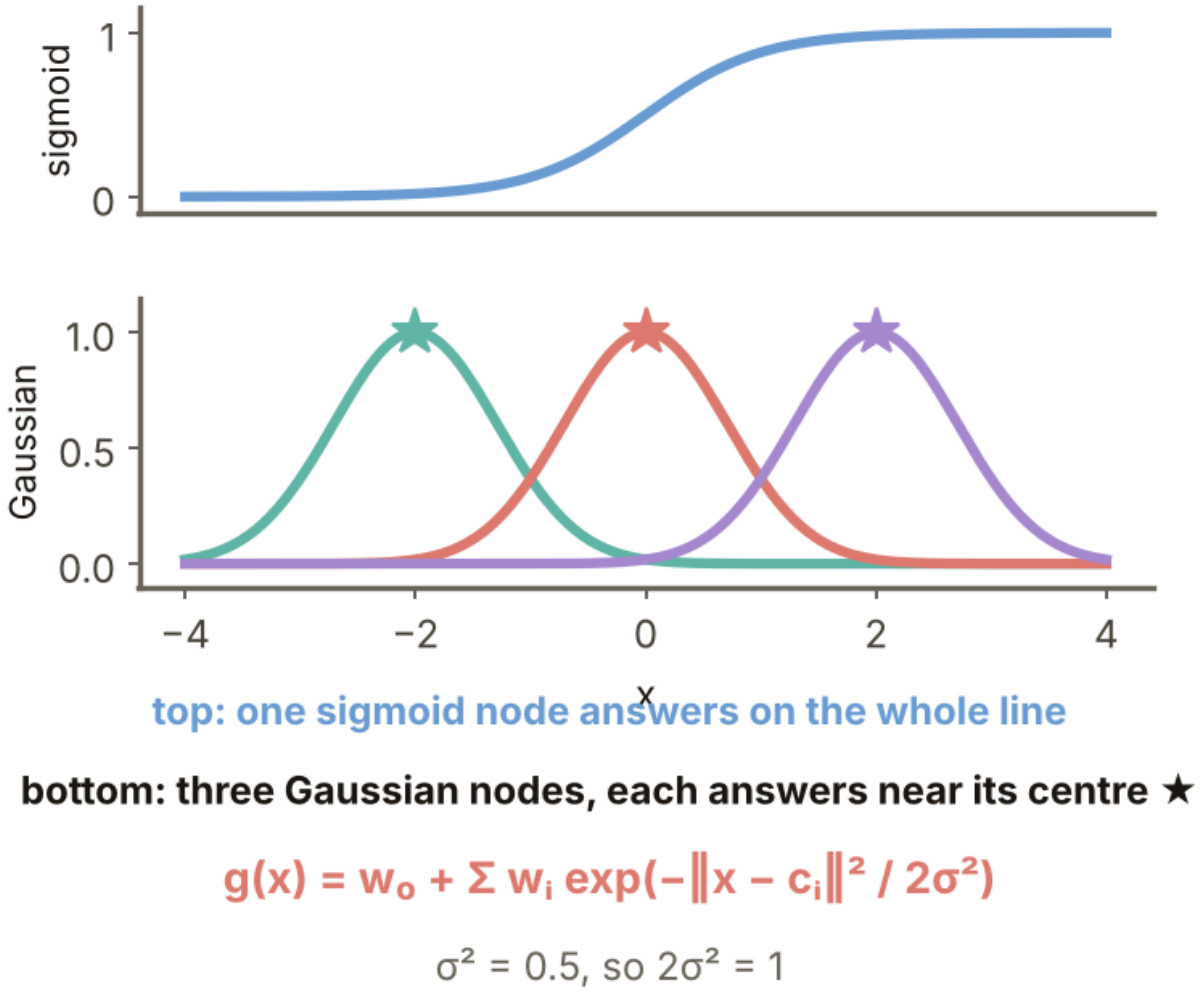
- **Radial basis function** (RBF): a function of the distance to a centre c_i, most often Gaussian.

Formal definition (the RBF network):

$$g(x) = w_0 + \sum_{i=1}^{k} w_i \exp\Big(- \frac{\|x - c_i\|^2}{2\sigma_i^2} \Big),$$

one layer of Gaussian hidden nodes and a linear output. It is close kin of the Parzen estimate, almost the same form, but Parzen uses one kernel per training point ($k = N$) while here $k \ll N$.

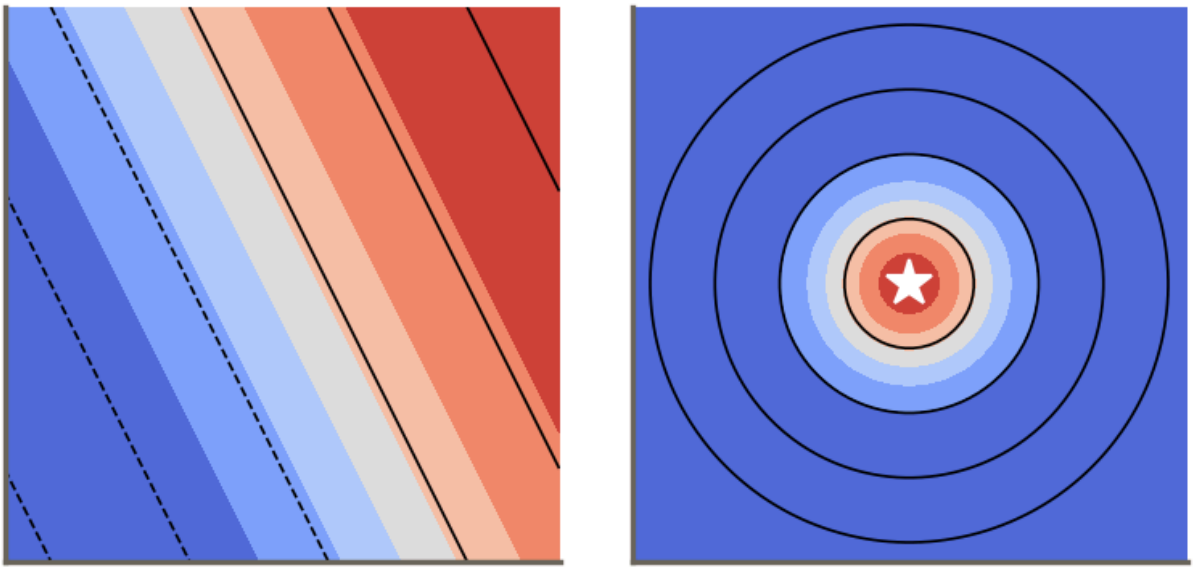
Figure `lec05-kp16-bumps`: top, one sigmoid answers along the whole line; bottom, three Gaussians each answer near their centre.



2. Local against global

Local response: a node answers only near its centre and fades to 0 away from it; a **global response** instead follows a whole hyperplane. The equal-output set of a sigmoid node is a hyperplane (a line in two dimensions); that of an RBF node is a hypersphere (a circle). **Example** ($2\sigma^2 = 1$): at distances 0, 1, 2, 3 from the centre the Gaussian answers $1, 0.368, 0.0183, 0.000123$.

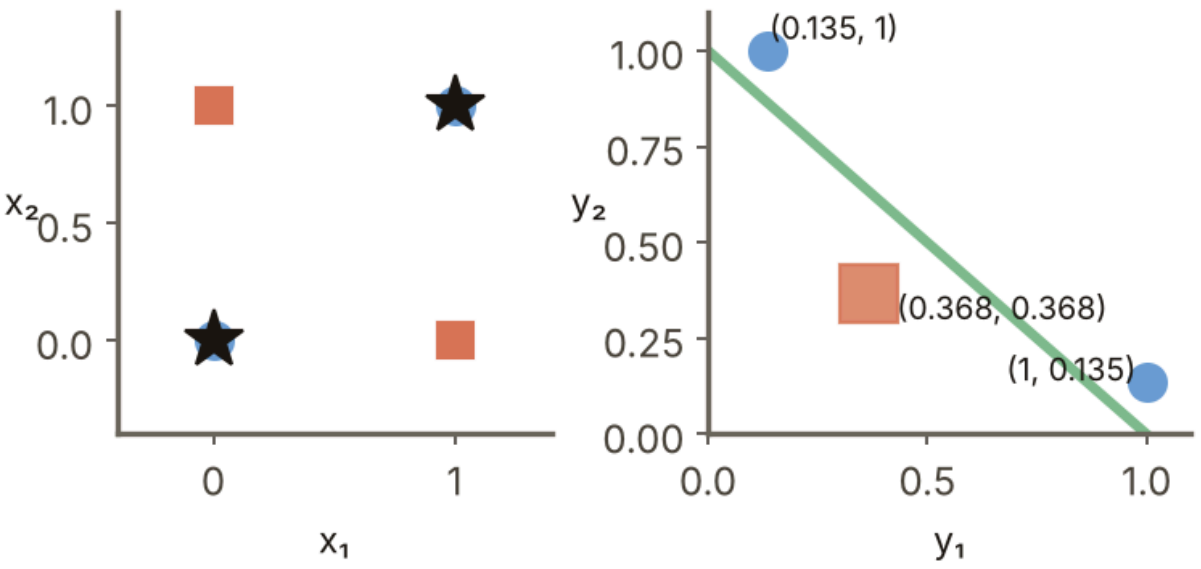
Figure `lec05-kp16-contours`: left the equal-output lines of a sigmoid, right the concentric circles of an RBF.



3. XOR needs only two centres

$k = 2$, $c_1 = (1, 1)$, $c_2 = (0, 0)$ and $f(x) = \exp(-\|x - c\|^2)$: $(0, 0) \to (0.135, 1)$, $(1, 1) \to (1, 0.135)$, and both $(1, 0)$ and $(0, 1)$ go to $(0.368, 0.368)$, folded onto one point again. $g(y) = y_1 + y_2 - 1$ gives $+0.135$ for the first two points and -0.264 for the other two, so one line separates them.

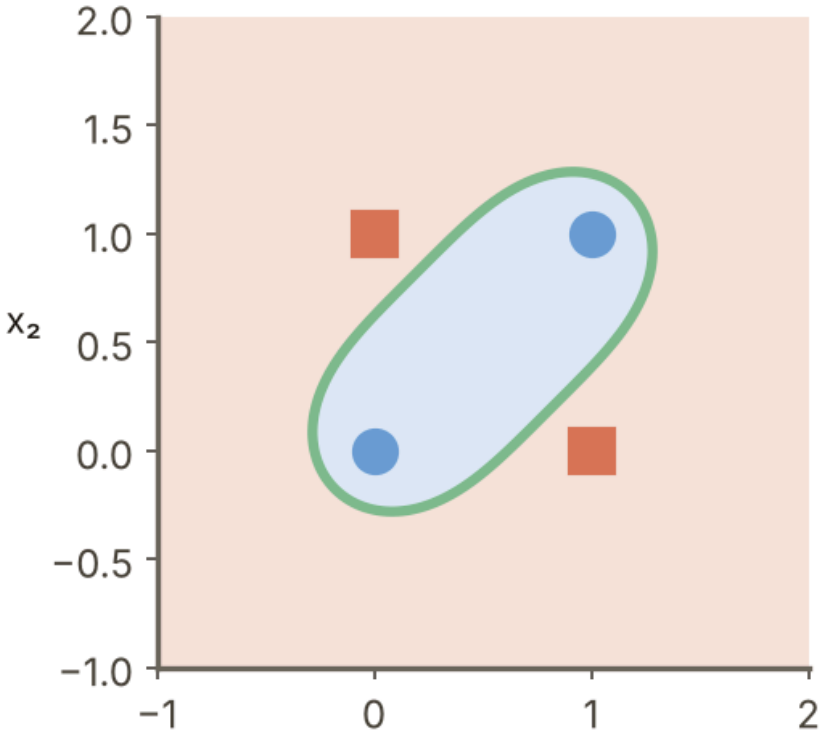
Figure `lec05-kp16-xormap`: the input plane (centres as stars) and the feature plane after the map.



4. Back in the original space

$g(x) = e^{-\|x-c_1\|^2} + e^{-\|x-c_2\|^2} - 1 = 0$ is a closed curve around the two centres. The values of g: $+0.135$ at $(0,0)$ and $(1,1)$ (inside the curve) and -0.264 at $(1,0)$ and $(0,1)$ (outside). Far from every centre $g \to -1$, so faraway points all fall on the outside class, a property of local response.

Figure `lec05-kp16-xorcurve`: the decision curve and the four points on the original plane.



$g(x) = e^{(-\|x-c_1\|^2)} + e^{(-\|x-c_2\|^2)} - 1 = 0$

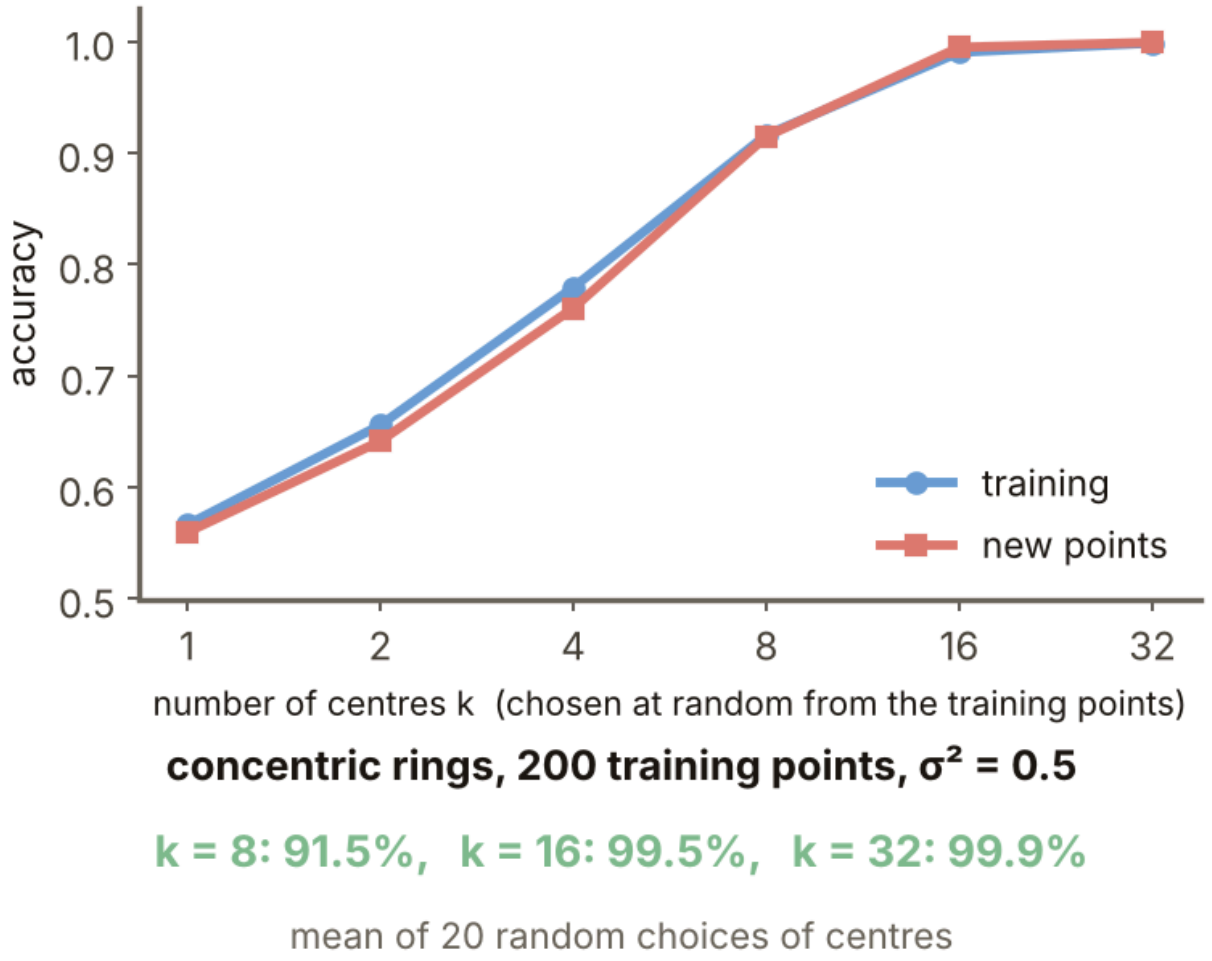
g at the four points: +0.135, +0.135, −0.264, −0.264

far from both centres g → −1: the default side

5. Fixed centres

Fixed centres: pick k training points at random as the centres; with the centres fixed only w remains, the linear problem of KP13 (least squares). **Example** (designed here): concentric rings (radii about 1 and 2.2), 200 training and 500 held-out new points, $\sigma^2 = 0.5$, averaged over 20 random choices for each k. Accuracy on new points: $k = 2$ gives 64.2%, $k = 8$ gives 91.5%, $k = 16$ gives 99.5% and $k = 32$ gives 99.9%.

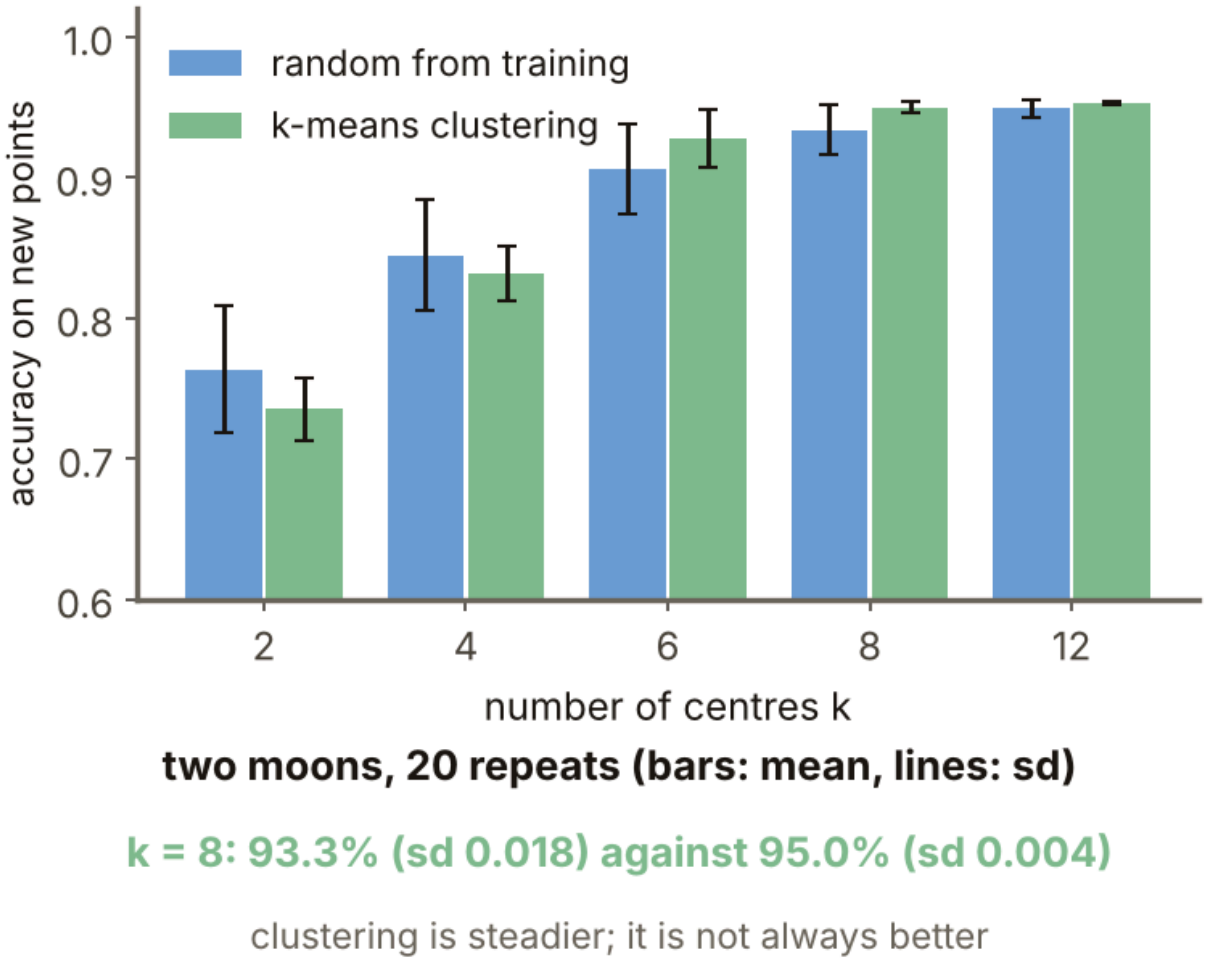
Figure `lec05-kp16-ringsk`: accuracy against the number of centres k.



6. Choosing the centres

Clustering (k-means here): split the data into k groups, use each group mean as a centre and then learn w with supervised least squares, a mix of unsupervised and supervised learning. **Example** (designed here): two moons of 60 points each with noise, 500 new points, $\sigma^2 = 0.3$, 20 repeats each. Accuracy on new points: $k = 6$: 90.5% against 92.7%; $k = 8$: 93.3% (sd 0.018) against 95.0% (0.004); at $k = 2, 4$ random is slightly higher. **Common misconception**: clustering is always more accurate than random picks. It is mainly steadier.

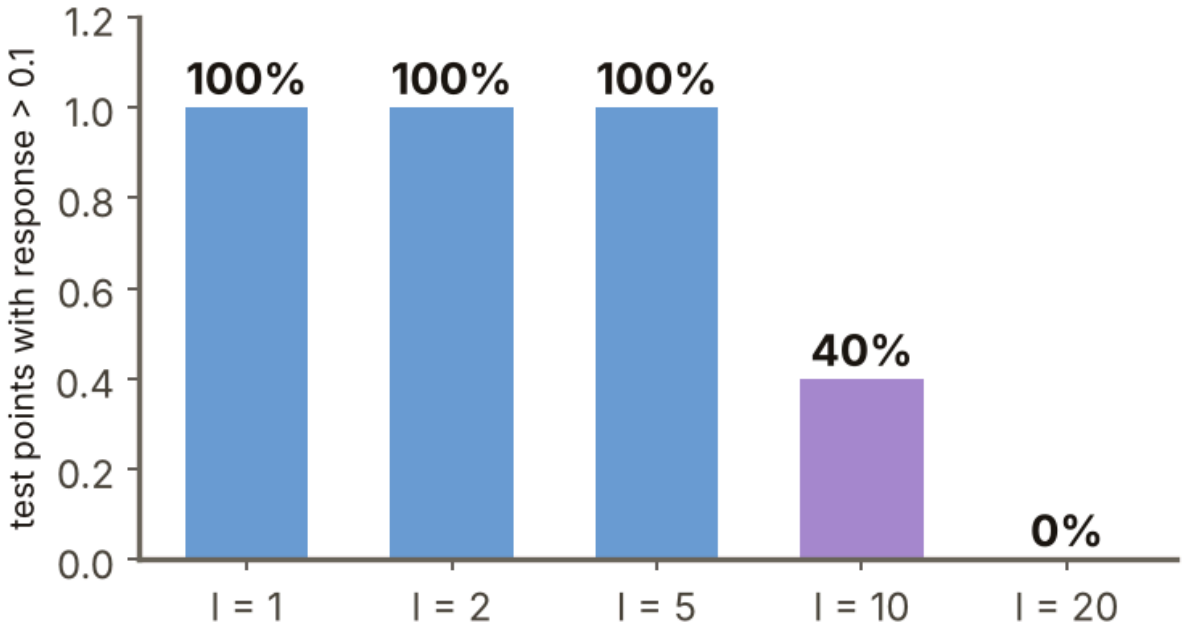
Figure `lec05-kp16-centers`: accuracy on new points for random and k-means centres.



7. The price of locality

Curse of dimensionality: as the dimension grows the number of points or centres needed to fill the space grows exponentially, while the data and the centres are never enough. **Example** (designed here): l is the input dimension; 100 random centres in the unit cube, $\sigma = 0.3$ and 2000 test points; the median distance to the nearest centre for $l = 2, 5, 10, 20$ is $0.05, 0.30, 0.67, 1.22$. Test points with a response above 0.1: 100% for $l \leq 5$, 40% for $l = 10$ and 0% for $l = 20$. A grid of 5 centres per axis needs 25, 3125 and 9,765,625 centres ($l = 2, 5, 10$).

Figure `lec05-kp16-curse`: the fraction of test points with a response above 0.1.



100 random centres in the unit cube, σ = 0.3

median nearest-centre distance: 0.05, 0.30, 0.67, 1.22 for l = 2, 5, 10, 20

a 5-per-axis grid needs 25, 3125, 9,765,625 centres (l = 2, 5, 10)

8. MLP against RBF

The MLP node is $\sum_j w_j x_j$ with hyperplanes and a global response; the RBF is $\|x - c_i\|$ with hyperspheres and a local one. The RBF learns faster; the MLP generalizes better where training data are sparse; for the same accuracy the RBF needs many more nodes (source). **To remember**: the MLP thinks in terms of which side of a hyperplane, the RBF in terms of how close to a centre; local learns fast but is hungry for dimensions.

Figure `lec05-kp16-compare`: comparison table of the MLP and the RBF network.

	MLP	RBF network
node argument	$\Sigma\, w_j x_j$	$\lVert x - c_i \rVert$
equal-output set	a hyperplane	a hypersphere
response	global	local
learning speed	slower	faster
sparse-data generalization	better	worse
nodes for the same accuracy	fewer	many more

MLP: which side of a hyperplane; RBF: how close to a centre

Extension: other ways, practice and worked solution

Other ways to choose the centres (source): train the centres (w_i, c_i, σ_i all by gradient descent, costly); orthogonal least squares (start from many candidates and pick a compact set by forward regression, which also estimates k); support vectors (view the RBF network as a kernel map; training is a quadratic programme with a global optimum and the number of centres decided automatically); growing (a new sample far from every centre with a large error becomes a new centre).

Figure `lec05-kp16-ways`: the six ways to choose the centres.

fixed centres

pick k training points at random; then only weights remain, a linear problem

train the centres

learn w, c, σ together by gradient descent; costly

clustering

cluster the data, centres = cluster means, then learn w

orthogonal least squares

pick a compact set from many candidates by forward regression

support vectors

quadratic programme; the centres are the support vectors

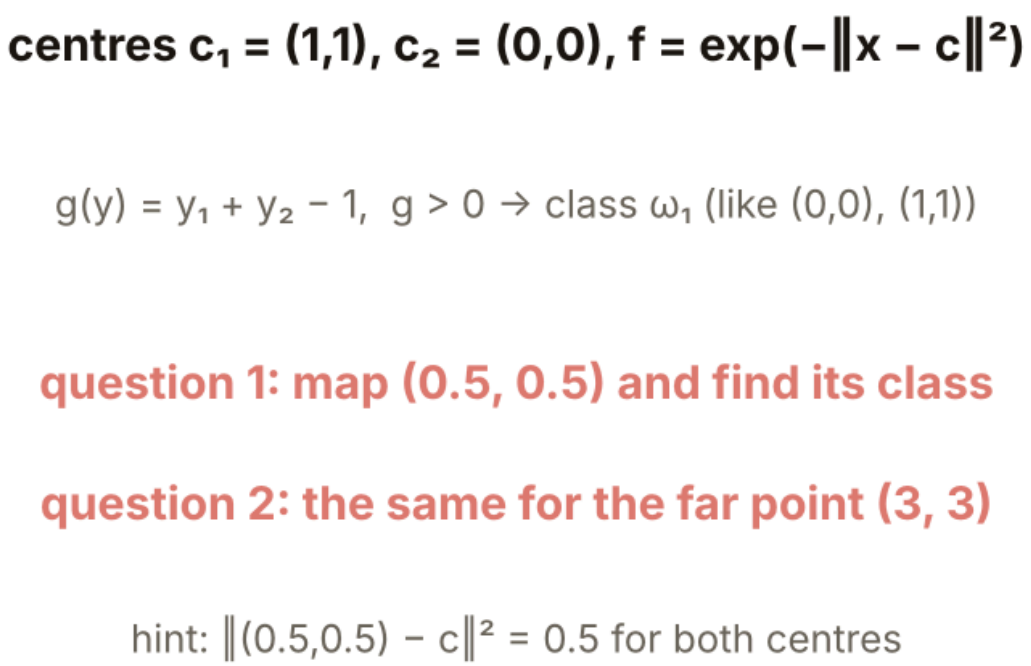
growing

add a centre when a sample is far from all and the error is large

Practice: $c_1 = (1, 1)$, $c_2 = (0, 0)$, $f = \exp(-\|x - c\|^2)$ and $g = y_1 + y_2 - 1$ ($g > 0$ is ω_1).

1. Map $(0.5, 0.5)$ to y and find its class. Hint: its squared distance to both centres is 0.5.
2. Do the same for the faraway point $(3, 3)$.

Figure `lec05-kp16-practice`: the setup of the practice.



Worked solution of questions 1 and 2: (1) $y = (e^{-0.5}, e^{-0.5}) = (0.607, 0.607)$ and $g = 0.607 + 0.607 - 1 = +0.213 > 0$, so ω_1. (2) The squared distances 8 and 18 give $y = (0.0003, 1.5 \times 10^{-8})$ and $g \approx -1.0 < 0$, so ω_2. (3) Check: the middle point lies between the centres, inside the curve; the faraway point falls to the bias -1 and takes the default class.

Figure `lec05-kp16-practice-sol`: the maps and g of the two points.

$(0.5, 0.5)$: $d^2 = 0.5, 0.5$	$y = (0.607, 0.607)$
$g = 0.607 + 0.607 - 1$	$= +0.213 \rightarrow \omega_1$
$(3, 3)$: $d^2 = 8, 18$	$y = (0.0003, 0.000000015)$
$g = 0.0003 + 0 - 1$	$\approx -1.0 \rightarrow \omega_2$

far from every centre the output falls to the bias -1

Terms and sources

English	中文
Radial basis function	徑向基函數
Local response	局部反應
Fixed centres	固定中心
Clustering	分群
Curse of dimensionality	維度詛咒

Source: Chapter 4 KP16, §4.15 (read 2026-10-25). The RBF form, its kinship with Parzen, the local-against-global comparison table, the two centres for XOR with 0.135 and 0.368, the centre-selection methods and the "to remember" line follow the source; the experiments on the number of centres for rings and moons, the random against k-means comparison and the dimension experiment on locality were designed and recomputed for this lecture ([added], (program)).